Chapter 6
Concepts of Macroeconomics

Syllabus topics covered

Growth; Unemployment; Inflation; Philips Curve; Business cycle; Circular flow of economics; Two-, three- and four-sector economics

Recommended books

1. K. K. Dewett: *Modern Economic Theory*, S. Chand Publishers
2. H. L. Ahuja: *Advanced Economic Theory*, S. Chand Publishers
3. Bangladesh Bank, BBS, NBR, Planning Commission and the Ministry of Finance.

6.1 Introduction to Macroeconomics

Macroeconomics is the branch of economics that studies the economy as a whole. It asks how the **total** output, employment, price level, income and growth of a nation are determined and how they change over time. The word comes from the Greek *makros* (large). The name was popularised by Ragnar Frisch in 1933, while the modern subject was founded by **J. M. Keynes** in *The General Theory of Employment, Interest and Money* (1936), written in response to the Great Depression.

Main objectives (goals) of macroeconomics

- **Economic growth** – steady rise in real output or GDP and per-capita income.
- **Full employment** – low unemployment and decent jobs.
- **Price stability** – low and predictable inflation.
- **External balance** – sustainable balance of payments and a stable exchange rate.
- **Equitable distribution** – reduction of poverty and inequality.

The Importance of Macroeconomics

- **Designing Government Budgets:** Helps policymakers allocate public funds efficiently, manage national debt, and structure taxation to stimulate growth.
- **Setting Monetary Policy & Interest Rates:** Guides central banks (like **Bangladesh Bank**) in fixing interest rates to balance borrowing costs for businesses while maintaining financial stability.
- **Controlling Inflation:** Provides the tools necessary to stabilize prices, which protects the purchasing power of citizens and maintains market confidence.
- **Planning Economic Growth:** Crucial for developing nations to map out long-term infrastructure, trade, and industrial strategies to escape poverty traps.
- **Creating Jobs for a Young Labor Force:** Helps design policies that foster private sector investment and industrialization, ensuring sustainable employment for a rapidly growing youth demographic.

Limitations

- Aggregates hide distribution: a rise in GDP may not reach the poor.
- Data are often incomplete, especially in the informal sector.
- Models rest on simplifying assumptions; policy lags make timing difficult.

6.2 Economic Growth

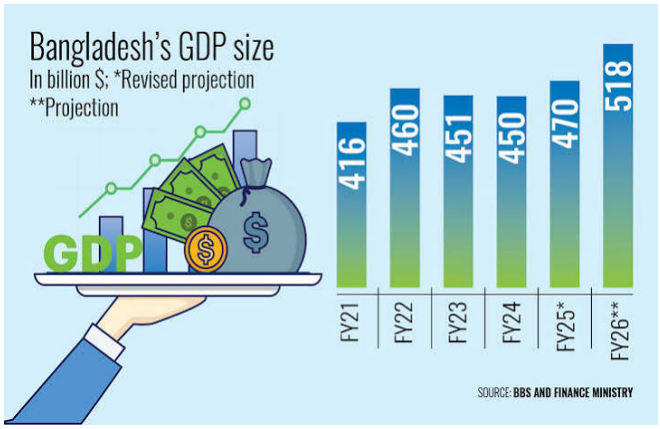
Economic growth is a sustained increase in a country's real output of goods and services over time, usually measured by the annual percentage change in **real GDP** or in **real GDP per capita**. Economist Kuznets defined it as "a long-term rise in capacity to supply increasingly diverse economic goods to its population".

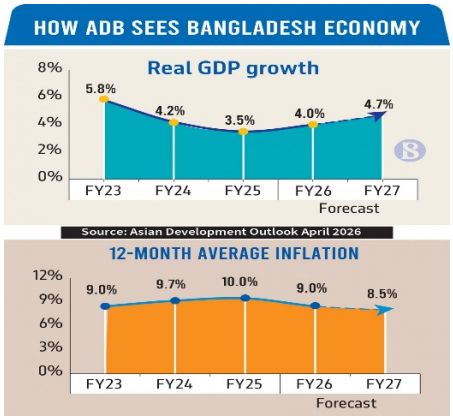
Growth rate (g) = [(Real GDP$_t$ − Real GDP$_{t-1}$) ÷ Real GDP$_{t-1}$] × 100

Example: if real GDP rises from Tk 40 lakh crore to Tk 42.4 lakh crore, g = (2.4 ÷ 40) × 100 = **6 per cent**. Per-capita growth is approximately real GDP growth minus population growth (e.g. 6% − 1.2% ≈ 4.8%).

Bangladesh context

- Bangladesh grew at roughly **6–7 per cent a year** for much of the **2010s** and became one of Asia's fastest-growing economies, driven by readymade garments, remittances, agriculture and rising domestic demand.
- Growth slowed after FY2022 amid foreign-exchange pressure, high inflation, energy shortages and political change; BBS provisional estimates put **FY2025** growth at about **4 per cent,** far below the Eighth Five-Year Plan target of **8.51** per cent for that year.
- The **FY2026–27** national budget targets **6.5 per cent** GDP growth and **7.5 per cent inflation**.





6.2 Economic Development

Economic development is a social process that improves the overall quality of life, economic well-being, and living standards of a nation.

Economic development is a multidimensional process through which a society improves the quality of life and well-being of its people by increasing income and employment, reducing poverty and inequality, transforming social and economic institutions, and expanding people's capabilities and freedoms, while ensuring the core values of development—sustenance, self-esteem, and freedom from servitude.

Economic growth versus economic development

Basis	Economic growth	Economic development
Meaning	Quantitative rise in output	Qualitative and quantitative improvement in living standards
Scope	Narrow: GDP and per-capita income	Broad: health, education, poverty, inequality, freedom, environment
Measure	Real GDP growth rate	Human Development Index, poverty and Gini measures
Nature	Can occur without structural change	Involves structural change in economy and society
Relevant for	All economies	Mainly developing economies
Question	**Economic Growth** answers the question **"how much?"** It tracks short-term quantitative expansions in output, income, and total jobs (measured by GDP or GNP).	**Economic Development** answers the question **"how well?"** It assesses long-term qualitative improvements in livability, sustainability, health, and institutional strength.

6.3 Unemployment

Definition and measurement

According to the ILO, a person is **unemployed** if he or she is without work, available for work and actively seeking work. The **labour force** consists of the employed plus the unemployed. People who are not seeking work (students, homemakers, retirees, discouraged workers) are outside the labour force.

Fig. 6.2 Population, Labour Force and Unemployment

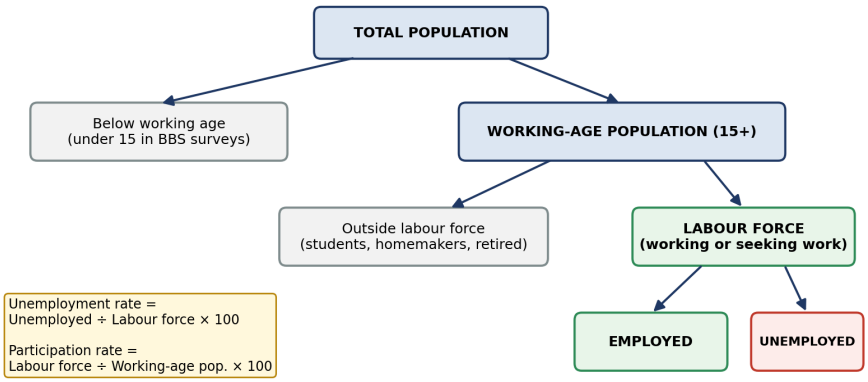


Fig. 6.2 Population, labour force and unemployment

Unemployment rate = (Number unemployed ÷ Labour force) × 100

Example: labour force = 7.5 crore, unemployed = 0.27 crore → unemployment rate = 0.27 ÷ 7.5 × 100 = **3.6 per cent**.

Types of unemployment

Type	Meaning	Example / cause
Frictional	Short-term joblessness while changing or searching for jobs	Fresh graduates searching for first job; imperfect information
Structural	Mismatch between workers' skills/location and available jobs	Skills gap between graduates and industry demand; decline of an industry
Cyclical (Keynesian)	Caused by deficient aggregate demand in recession	Factory layoffs during a downturn
Seasonal	Jobs available only in certain seasons	Agricultural workers between planting and harvest; "monga" in north Bengal
Technological	Machines replace labour	Automation in textile mills
Disguised (hidden)	Workers whose marginal product is zero or very low; removing them does not reduce output	Excess family labour on small farms in an over-populated rural economy
Underemployment	Working fewer hours or below skill level than desired	Graduate working as a delivery rider
Classical (real-wage)	Real wages set above market-clearing level	Minimum wages above productivity
Voluntary vs involuntary	Voluntary: refuse available work at going wage. Involuntary: willing but cannot find work	Keynes was concerned with involuntary unemployment

Bangladesh context

- The headline unemployment rate reported in BBS Labour Force Surveys is low (around 3–4 per cent) because the ILO definition counts even one hour of work as employment.
- The real problems are **underemployment, informal work (about 85 per cent of jobs)**, low productivity and high unemployment among educated youth; the Eighth Five-Year Plan set a target of creating about 11.6 million jobs (8.17 million at home and about 3.5 million abroad).

6.4 Inflation

Definition

Inflation is a sustained rise in the general level of prices of goods and services over time, which means a fall in the purchasing power of money. A one-time rise in the price of a single good is not inflation. Economists describe it as "too much money chasing too few goods" (Coulborn).

Measurement

Index	What it measures	Remarks
Consumer Price Index (CPI)	Cost of a fixed basket of goods and services bought by households	Most common measure; BBS publishes monthly CPI (national, urban, rural) and point-to-point and 12-month average inflation
GDP deflator	(Nominal GDP ÷ Real GDP) × 100; covers all domestically produced goods	Not a fixed basket; includes investment and government goods
Wholesale Price Index (WPI)	Prices at the wholesale/producer level	Often leads CPI; used in some countries
Core inflation	CPI excluding volatile items (food and energy)	Shows underlying trend

Inflation rate = $[(CPI_t - CPI_{t-1}) \div CPI_{t-1}] \times 100$

Example: CPI rises from 120 to 126.6 over a year → inflation = (6.6 ÷ 120) × 100 = **5.5 per cent**.

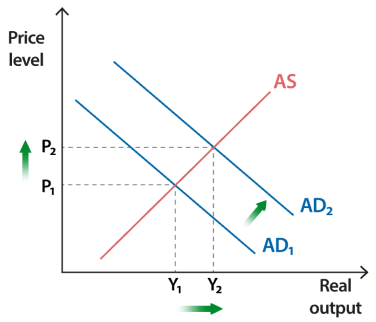
Causes of inflation

The main causes of inflation can be grouped into 2 broad categories:

1. **demand-pull,**
2. **cost-push,**

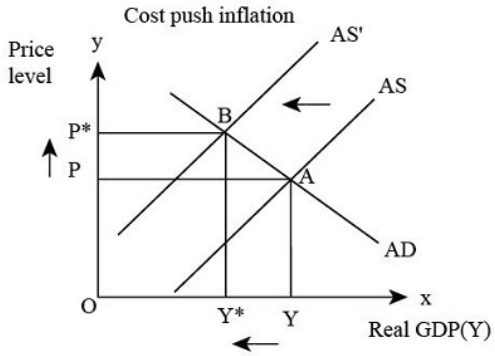
Demand-pull inflation

Demand-pull inflation arises when the total demand for goods and services (i.e. 'aggregate demand') increases to exceed the supply of goods and services (i.e. 'aggregate supply') that can be sustainably produced. The excess demand puts upward pressure on prices across a broad range of goods and services and ultimately leads to an increase in inflation – that is, it 'pulls' inflation higher.



Cost-push inflation

Cost-push inflation occurs when the total supply of goods and services in the economy which can be produced (aggregate supply) falls. A fall in aggregate supply is often caused by an increase in the cost of production. If aggregate supply falls but aggregate demand remains unchanged, there is upward pressure on prices and inflation – that is, inflation is 'pushed' higher.



Bangladesh context

- BBS-reported inflation stayed high, around **9–10 per cent** during FY2023–FY2025, driven by food prices, imported-cost pressure from taka depreciation, energy price adjustments and supply-chain problems.
- The FY2026–27 budget aims to bring inflation down to **7.5 per cent**. Bangladesh Bank follows a tight monetary stance and has moved toward a more market-based exchange rate and interest-rate framework.

6.5 The Phillips Curve

In 1958 the New Zealand economist **A. W. Phillips** studied UK data for 1861–1957 and found a stable **inverse relation between the rate of unemployment and the rate of change of money wages**. Samuelson and Solow (1960) restated it as a trade-off between **unemployment and inflation**: low unemployment is associated with high inflation, and vice versa.

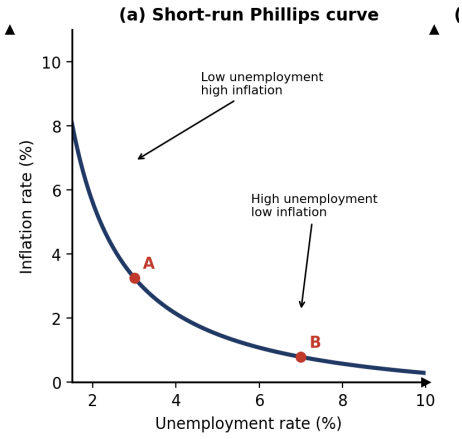


Fig. 6.4 Short-run Phillips curve and the long-run (vertical) Phillips curve

Why does the trade-off exist? (Short run)

- When aggregate demand rises, firms hire more workers; unemployment falls and labour becomes scarce.
- Wages and then prices are bid up, so inflation rises. The curve is convex (steeper at low unemployment) because labour shortages push wages up faster.

- Policy-makers face a **menu of choices**: point A (low unemployment, high inflation) or B (high unemployment, low inflation).

6.6 The Business Cycle

A **business cycle** (trade cycle) is the recurrent, wave-like fluctuation of aggregate economic activity (real GDP, employment, investment, prices) around its long-run growth trend. Burns and Mitchell (NBER) described cycles as expansions occurring at about the same time in many economic activities, followed by general recessions, contractions and revivals which merge into the next expansion; the sequence is recurrent but **not periodic**.

Fig. 6.5 Phases of the Business Cycle

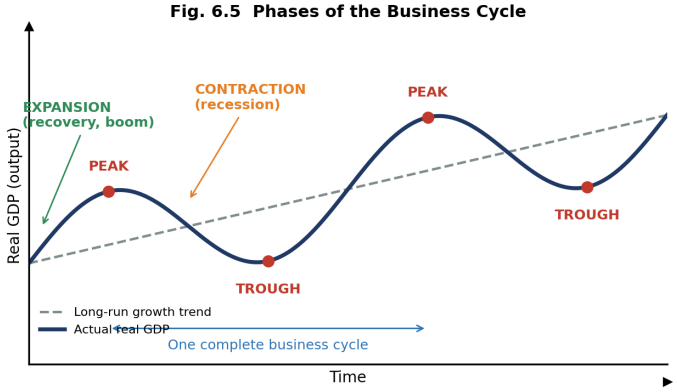


Fig. 6.5 Phases of the business cycle

Phases

Phase	Features
Expansion (recovery / prosperity)	Rising output, employment, investment, profits and prices; credit expands; business confidence high
Peak (boom)	Output at maximum capacity; inflation high; costs rise and profit margins squeeze
Contraction (recession)	Demand falls, investment and output decline, layoffs begin; two consecutive quarters of negative real GDP growth is the common rule of thumb for a recession
Trough (depression)	Output and employment at lowest; deflation or low inflation; excess capacity; confidence low. Revival begins when stocks are exhausted and low interest rates stimulate investment

Bangladesh context

Bangladesh is a developing economy with a large agricultural and informal sector, so its cycles are shaped by monsoon and flood shocks, export demand from Europe and the USA, remittance inflows and global commodity prices. The COVID-19 shock (FY2020) and the 2022–24 external-sector stress are recent examples of downturns.

6.7 Circular Flow of Economic Activity

The **circular flow** model shows how income, output and expenditure continuously flow between the sectors of an economy. Production creates income for factor owners; income is spent on goods and services; spending becomes revenue for firms and pays for new production. The cycle is therefore **self-perpetuating**.

Real flow and money flow

- **Real flow:** flow of factor services (land, labour, capital, enterprise) from households to firms and goods and services from firms to households.
- **Money flow:** factor payments (wages, rent, interest, profit) flow from firms to households, and consumption expenditure flows from households to firms. Money flows are opposite to real flows.

Leakages and injections

Leakages are withdrawals from the circular flow (saving S, taxes T, imports M). **Injections** are additions to the flow (investment I, government spending G, exports X). Income remains stable when total injections equal total leakages.

6.7.1 Two-sector economy (households and firms)

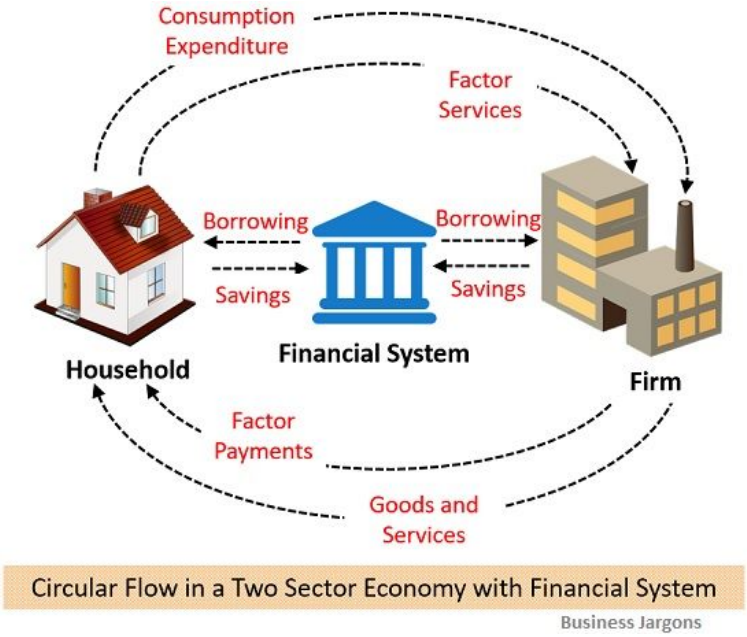


Fig. 6.6 Two-sector circular flow

- **Assumptions:** closed economy, no government; households spend all income (no saving) in the simplest case.
- With saving, a leakage appears; **financial markets** channel S to firms for investment I.
- **Equilibrium:** Y = C + I and equivalently S = I.

6.7.2 Three-sector economy (households, firms, government)

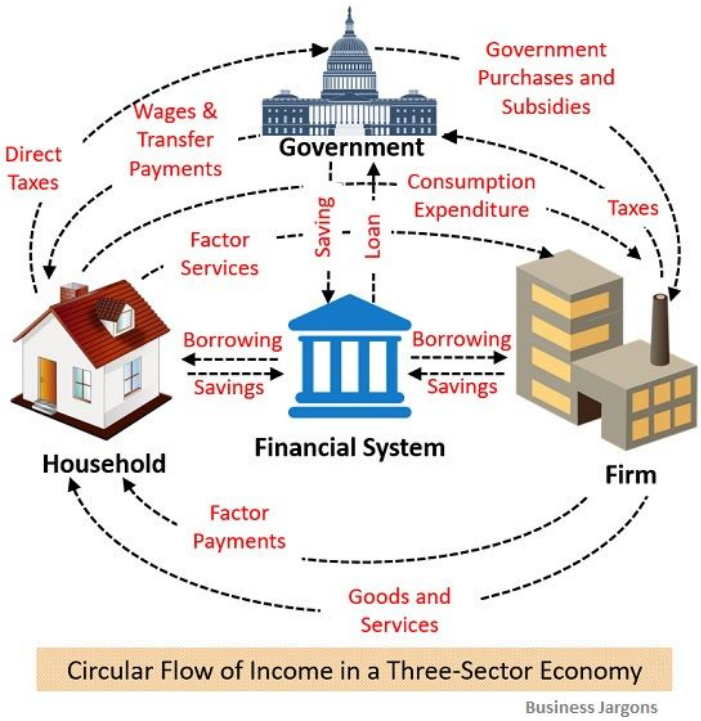


Fig. 6.7 Three-sector circular flow

- Government collects taxes T (leakage) and spends G on goods, services and transfers (injection).
- **Equilibrium:** Y = C + I + G and S + T = I + G.

6.7.3 Four-sector economy (open economy)

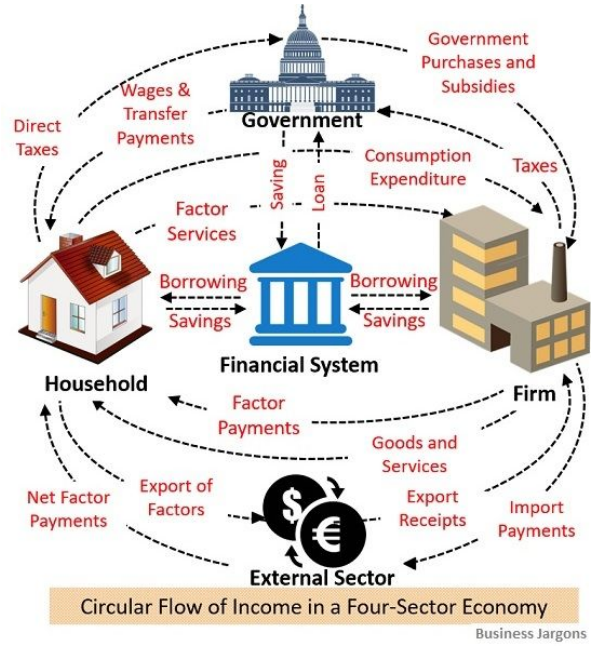


Fig. 6.8 Four-sector circular flow

- Foreign trade adds exports X (injection) and imports M (leakage); net exports = X − M.
- **Equilibrium:** Y = C + I + G + (X − M) and S + T + M = I + G + X.

6.8 Two-, Three- and Four-Sector Economics

Summary table

	Two-sector	Three-sector	Four-sector
Sectors	Households, firms	+ Government	+ Rest of the world
Aggregate expenditure	C + I	C + I + G	C + I + G + (X − M)
Leakages	S	S + T	S + T + M
Injections	I	I + G	I + G + X
Equilibrium	S = I	S + T = I + G	S + T + M = I + G + X

Broad questions

1. Explain the types, causes and remedies of unemployment, with reference to Bangladesh.
2. Discuss the causes and effects of inflation and the measures to control it.
3. Explain the Phillips curve. How did Friedman and Phelps modify it?
4. Discuss the main theories of the business cycle and the policy measures to control it.
5. Explain the circular flow of income in two-, three- and four-sector economies using diagrams.